

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
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Section / Batch:	AIML	Date:	22/07/2026

Code of Conduct

I K. Srinivasa Reddy (1925254452) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: K. Srinivasa Reddy

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25

Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int n, int key) {  
4     int low = 0, high = n - 1;  
5     while (low <= high) {  
6         int mid = (low + high) / 2;  
7         if (arr[mid] == key)  
8             return mid;  
9         else if (arr[mid] < key)  
10            low = mid + 1;  
11        else  
12            high = mid - 1;  
13    }  
14    return -1;  
15 }  
16  
17 int main() {  
18  
19     int arr[] = {1, 3, 5, 7, 9, 11, 13};  
20     int n = sizeof(arr) / sizeof(arr[0]);  
21     int key = 7;  
22  
23  
24     int idx = binarySearch(arr, n, key);  
25     if (idx != -1)  
26         printf("Output Element found at index %d\n", idx);  
27     else  
28         printf("Output Element not found\n");  
29  
30     return 0;  
31 }
```

Test Input

```
arr={1,3,5,7,9,11,13}  
n=7  
key=7
```

Output

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 int pop() {  
9     if (top == -1) {  
10         printf("Underflow");  
11         return -1;  
12     }  
13     return stack[top--];  
14 }  
15  
16 int main() {  
17     int n, i, x;  
18     scanf("%d", &n);  
19     for (i = 0; i < n; i++) {  
20         scanf("%d", &x);  
21         stack[++top] = x;  
22     }  
23     for (i = 0; i < n; i++) {  
24         printf("%d ", pop());  
25     }  
26     return 0;  
27 }
```

Test Input

```
3  
10 20 30
```

Output

 Run Save Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List Data Structures
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init() {
7     top = -1; // Fix: start at -1 for an empty stack
8 }
9
10 void push(int val) {
11     if (top < 4) stack[++top] = val;
12 }
13
14 int pop() {
15     if (top >= 0) return stack[top--];
16     return -1;
17 }
18
19 int main() {
20     init();
21     push(10);
22     push(20);
23     printf("%d\n", pop());
24     printf("%d\n", pop());
25     return 0;
26 }
```

Test Input

Output

[Run](#)[Save](#)[Submit Fix](#)

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ' '){
    while(stack[top] != ' '){
        postfix[j++] = pop();
    }
    pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 char stack[100];
4 int top = -1;
5
6 void push(char x) { stack[++top] = x; }
7 char pop() { return stack[top--]; }
8 char peek() { return stack[top]; }
9
10 int main() {
11     char infix[] = "(a+b)*c"; // Example input
12     char postfix[100];
13     int j = 0;
14
15     for (int i = 0; infix[i] != '\0'; i++) {
16         char ch = infix[i];
17
18         if (ch >= 'a' && ch <= 'z') {
19             postfix[j++] = ch;
20         } else if (ch == '(') {
21             push(ch);
22         } else if (ch == ')') {
23             // The debugged part:
24             while (top != -1 && peek() != '(') {
25                 postfix[j++] = pop();
26             }
27             if (top != -1) pop(); // Pop '('
28         } else {
29             push(ch);
30         }
31     }
32
33     while (top != -1) postfix[j++] = pop();
34     postfix[j] = '\0';
35
36     printf("Postfix: %s\n", postfix);
37     return 0;
38 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||  
(ch == '[' && stack[top] == ']')) {  
    pop();  
} else {  
    return 0;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[20];  
4 int top = -1;  
5  
6 char pop() {  
7     return stack[top--];  
8 }  
9  
10 int check(char ch) {  
11     if (top == -1)  
12         return 0;  
13  
14     if ((ch == ']' && stack[top] == '[') ||  
15         (ch == '[' && stack[top] == ']') ||  
16         (ch == '[' && stack[top] == '(') {  
17         pop();  
18         return 1;  
19     }  
20  
21     return 0;  
22 }  
23  
24 int main() {  
25     stack[++top] = '[';  
26  
27     if (check('['))  
28         printf("Bracket matched");  
29     else  
30         printf("Bracket not matched");  
31  
32     return 0;  
33 }
```

📌 Submitted

Test Input

[]

Output

Bracket matched

⏸ Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List Data Structures
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q6 Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7
8 int main() {
9     struct Node n1, n2, n3;
10    struct Node *head = &n1;
11    struct Node *temp;
12
13    n1.data = 10;
14    n1.next = &n2;
15
16    n2.data = 20;
17    n2.next = &n3;
18
19    n3.data = 30;
20    n3.next = NULL;
21
22    temp = head;
23
24    while (temp != NULL) {
25        printf("%d ", temp->data);
26        temp = temp->next;
27    }
28
29    return 0;
30 }
```

Test Input

stdin input (optional)

Output

10 20 30

⚠ Pending manual review

📌 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

```
POP(stack):  
if top == 0:  
    print("Underflow")  
else:  
    return stack[top--]
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[5] = {10, 20, 30};  
4 int top = 2;  
5  
6 int pop() {  
7     if (top == -1) {  
8         printf("Underflow\n");  
9         return -1;  
10    }  
11  
12    return stack[top--];  
13 }  
14  
15 int main() {  
16     printf("Deleted element = %d", pop());  
17     return 0;  
18 }
```

Submitted

Test Input

stdin input (optional)

Output

Deleted element = 30

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7
8 int main() {
9     struct Node n1, n2, n3;
10    struct Node *head, *current, *prev, *next;
11
12    n1.data = 10;
13    n1.next = &n2;
14
15    n2.data = 20;
16    n2.next = &n3;
17
18    n3.data = 30;
19    n3.next = NULL;
20
21    head = &n1;
22    current = head;
23    prev = NULL;
24
25    while (current != NULL) {
26        next = current->next;
27        current->next = prev;
28        prev = current;
29        current = next;
30    }
31
32    head = prev;
33
34    printf("Reversed List: ");
35
36    current = head;
37    while (current != NULL) {
38        printf("%d ", current->data);
39        current = current->next;
40    }
41
42    return 0;
43 }
```

Test Input

stdin input (optional)

Output

Reversed List: 30 20 10

⏱ Pending manual review

✔ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q9 Linked List

5 marks

What is the error in the below code?

new.next = head

head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *newNode;
12
13     newNode = (struct Node *)malloc(sizeof(struct Node));
14
15     newNode->data = 10;
16     newNode->next = head;
17     head = newNode;
18
19     printf("Inserted element = %d", head->data);
20
21     free(newNode);
22     return 0;
23 }
```

Submitted

Test Input

stdin input (optional)

Output

Inserted element = 10

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node {  
5     int data;  
6     struct Node *next;  
7 };  
8  
9 int main() {  
10     struct Node *head = NULL;  
11     struct Node *newNode;  
12  
13     newNode = (struct Node *)malloc(sizeof(struct Node));  
14  
15     newNode->data = 10;  
16     newNode->next = head;  
17     head = newNode;  
18  
19     printf("Inserted element = %d", head->data);  
20  
21     free(newNode);  
22  
23     return 0;  
24 }
```

Test Input

stdin input (optional)

Output

Inserted element = 10

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

10/10 answered

Q1 Stack Notations - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[10], top = -1;  
4  
5 void push(int x) {  
6     stack[++top] = x;  
7 }  
8  
9 int pop() {  
10    return stack[top--];  
11 }  
12  
13 int main() {  
14     char expression[] = "23+";  
15     int i = 0, result;  
16  
17     while (expression[i] != '\0') {  
18         if (expression[i] >= '0' && expression[i] <= '9')  
19             push(expression[i] - '0');  
20         else {  
21             int op2 = pop();  
22             int op1 = pop();  
23             result = op1 + op2;  
24             push(result);  
25         }  
26         i++;  
27     }  
28  
29     printf("Result = %d", pop());  
30  
31     return 0;  
32 }
```

Test Input

23+

Output

Result = 5

⚠ Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX] = {10, 20, 30};  
6 int front = 0, rear = 2;  
7  
8 int dequeue() {  
9     if (front == -1) {  
10         printf("Underflow\n");  
11         return -1;  
12     }  
13  
14     int data = queue[front];  
15  
16     if (front == rear)  
17         front = rear = -1;  
18     else  
19         front = (front + 1) % MAX;  
20  
21     return data;  
22 }  
23  
24 int main() {  
25     int x = dequeue();  
26  
27     if (x != -1)  
28         printf("Deleted element = %d", x);  
29  
30     return 0;  
31 }
```

Test Input

stdin input (optional)

Output

Deleted element = 10

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int l, int r, int x) {  
4     if (l > r)  
5         return -1;  
6  
7     int mid = l + (r - l) / 2;  
8  
9     if (arr[mid] == x)  
10        return mid;  
11  
12    if (arr[mid] > x)  
13        return binarySearch(arr, l, mid - 1, x);  
14  
15    return binarySearch(arr, mid + 1, r, x);  
16 }  
17  
18 int main() {  
19     int arr[] = {10, 20, 30, 40, 50};  
20     int x = 40;  
21  
22     int result = binarySearch(arr, 0, 4, x);  
23  
24     if (result != -1)  
25         printf("Element found at index %d", result);  
26     else  
27         printf("Element not found");  
28  
29     return 0;  
30 }
```

Submitted

Test Input

stdin input (optional)

Output

Element found at index 3

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q4 Queue

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int deque[MAX];  
6 int front = 2, rear = 2;  
7  
8 void insertFront(int data) {  
9     if ((front == 0 && rear == MAX - 1) ||  
10         front == rear + 1) {  
11         printf("Deque Full");  
12         return;  
13     }  
14  
15     if (front == 0)  
16         front = MAX - 1;  
17     else  
18         front = front - 1;  
19  
20     deque[front] = data;  
21 }  
22  
23 int main() {  
24     insertFront(10);  
25  
26     printf("Inserted element = %d", deque[front]);  
27  
28     return 0;  
29 }
```

Test Input

Front = 2
Data = 10

Output

Inserted element = 10

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix
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Circular Queue
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5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define MAX 5  
3  
4 int queue[MAX];  
5 int rear = 0;  
6  
7 void enqueue(int data) {  
8     if (rear == MAX) {  
9         printf("Queue Full\n");  
10        return;  
11    }  
12  
13    queue[rear++] = data;  
14 }  
15  
16 int main() {  
17     enqueue(10);  
18     enqueue(20);  
19  
20     printf("Queue: %d %d", queue[0], queue[1]);  
21  
22     return 0;  
23 }
```

Submitted

Test Input

stdin input (optional)

Output

Queue: 10 20

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QUESTIONS

Q1 ✓

Stack Notations - Postfix
Expression
5 marks

Q2 ✓

Circular Queue
5 marks

Q3 ✓

Binary Search
5 marks

Q4 ✓

Queue
5 marks

Q5 ✓

Queue
5 marks

Q6 ✓

Doubly Linked List
5 marks

Q7 ✓

Stack
5 marks

Q8 ✓

Stack
5 marks

Q9 ✓

Circular Queue
5 marks

Q10 ✓

Stack
5 marks

10/10 answered

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node {  
5     int data;  
6     struct Node *prev;  
7     struct Node *next;  
8 };  
9  
10 void deleteNode(struct Node **head, struct Node *del) {  
11     if (del == NULL)  
12         return;  
13  
14     if (*head == del)  
15         *head = del->next;  
16  
17     if (del->prev != NULL)  
18         del->prev->next = del->next;  
19  
20     if (del->next != NULL)  
21         del->next->prev = del->prev;  
22  
23     free(del);  
24 }  
25  
26 int main() {  
27     printf("Node deletion operation completed.");  
28  
29     return 0;  
30 }
```

Test Input

stdin input (optional)

Output

Node deletion operation completed.

Pending manual review

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QUESTIONS

Q1 ✓

Stack Notations - Postfix
Expression
5 marks

Q2 ✓

Circular Queue
5 marks

Q3 ✓

Binary Search
5 marks

Q4 ✓

Queue
5 marks

Q5 ✓

Queue
5 marks

Q6 ✓

Doubly Linked List
5 marks

Q7 ✓

Stack
5 marks

Q8 ✓

Stack
5 marks

Q9 ✓

Circular Queue
5 marks

Q10 ✓

Stack
5 marks

Q7 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow");
9         return -1;
10    }
11
12    return stack[top--];
13 }
14
15 int main() {
16     printf("Popped element = %d", pop());
17
18     return 0;
19 }
```

Submitted

Test Input

stdin input (optional)

Output

Popped element = 30

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

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Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

10/10 answered

Q8 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

🔍 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = MAX - 1;
7
8 int main() {
9     if (top == MAX - 1)
10         printf("Overflow");
11     else
12         printf("Space available");
13
14     return 0;
15 }
```

Test Input

stdin input (optional)

Output

Overflow

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 ✓

Stack Notations - Postfix
Expression
5 marks

Q2 ✓

Circular Queue
5 marks

Q3 ✓

Binary Search
5 marks

Q4 ✓

Queue
5 marks

Q5 ✓

Queue
5 marks

Q6 ✓

Doubly Linked List
5 marks

Q7 ✓

Stack
5 marks

Q8 ✓

Stack
5 marks

Q9 ✓

Circular Queue
5 marks

Q10 ✓

Stack
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;  
queue[rear] = vehicle;  
if(rear == front){  
    printf("Queue Full");  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = 0, rear = -1;  
7  
8 void enqueue(int vehicle) {  
9     if ((rear + 1) % MAX == front) {  
10         printf("Queue Full\n");  
11         return;  
12     }  
13  
14     rear = (rear + 1) % MAX;  
15     queue[rear] = vehicle;  
16  
17     printf("Vehicle inserted = %d\n", vehicle);  
18 }  
19  
20 int main() {  
21     enqueue(101);  
22  
23     return 0;  
24 }
```

Submitted

Test Input

stdin input (optional)

Output

Queue Full

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1**
Stack Notations - Postfix Expression
5 marks
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Doubly Linked List
5 marks
- Q7**
Stack
5 marks
- Q8**
Stack
5 marks
- Q9**
Circular Queue
5 marks
- Q10**
Stack
5 marks

10/10 answered

Q10 Stack

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int peek() {
7     if (top == -1) {
8         printf("Underflow");
9         return -1;
10    }
11    return stack[top];
12 }
13
14 int main() {
15     printf("Top element = %d", peek());
16     return 0;
17 }
```

Submitted

Test Input

stdin input (optional)

Output

Top element = 30

Pending manual review

My Assessments

Course: CSA0309RB

PSEUDOCODE WRITING TASK OPEN

COS_AT1_PSEUDOCODE WRITING TASK

0/5 answered

Attempt →

CO3_AT3_MCQ OPEN

Multiple Choice Questions

0/25 answered

Attempt →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN ✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

C CODE OUTPUT PREDICTION

10/10 answered

Review →